Connect Deeply

Product & Technical Plan

Person-intelligence app for conferences and meetings

Prepared July 7, 2026

Search a person before or during a conversation and get a synthesized dossier — career history, interests, socials, portfolio, research/patents, mutual connections — plus AI-generated common ground and icebreakers.

1. Product framing

Working name: **Connect Deeply**. Positioning: a mobile-first "pre-conversation intelligence" tool — search a name before or during a meeting/conference, get a synthesized profile plus AI-generated icebreakers, then save notes afterward (personal CRM).

Does this exist already?

Nothing does exactly this. The closest analogues are all B2B-sales-flavored, not conference/networking-flavored:

- **Humanlinker, Crystal Knows, Humantic AI, Signum.ai** — LinkedIn scraping + personality analysis + AI icebreakers, but built for sales prospecting, not general networking.
- **Apollo.io, Clearbit, People Data Labs, FullContact** — licensed contact/company enrichment APIs. These are the *data layer* to build on, not consumer competitors.
- **Brella, Swapcard, Grip** — event AI matchmaking, but based on self-submitted, opted-in profile data inside a closed event platform — a fundamentally different (and safer) data model than open-web aggregation.

Conclusion: there is a genuine product gap for a mobile-first, consumer-facing, cross-platform "who is this person and how do I open the conversation" app. The reason the gap exists is worth taking seriously — see the legal section below.

2. Data-sourcing decision & boundary

Decision made: pursue open scraping for maximum data depth, rather than licensed-API-only or a two-sided opt-in network.

Hard boundary: build the scraper to touch only logged-out, publicly rendered pages, at a conservative pace with real backoff. Do **not** build fake-account farms, credential-based/logged-in scraping, CAPTCHA-bypass, or IP-rotation systems purpose-built to evade a platform's anti-bot defenses. That exact pattern (fake accounts + gated-data scraping) is what got Proxycurl — the largest LinkedIn-scraping API on the market — sued by LinkedIn and shut down permanently on July 4, 2025. If deeper LinkedIn data is needed later, source it from a licensed provider, not evasion tooling.

3. Data source map

How each requested data point actually gets sourced, and its relative legal risk.

You want	Source	Method	Risk
LinkedIn profile, headline, work history	LinkedIn public profile page	Scraper (logged-out)	High
LinkedIn posts / activity	LinkedIn public activity page	Scraper (logged-out)	High
GitHub repos, languages, orgs	GitHub REST API	Official API	None

You want	Source	Method	Risk
Research papers	Semantic Scholar / OpenAlex	Official API	None
Patents	Google Patents / USPTO	Official API	None
Portfolio / personal site	Google Search + site fetch	Search + crawl of their own site	Low
Instagram	No public 3rd-party API	Scraper (logged-out only)	Highest — defer past MVP
Twitter/X handle, bio, posts	Limited official API tiers, else public page	API where affordable, else scraper	Medium
Mutual connections	"Sign in with LinkedIn" on your own users, cross-ref target	Official, but only covers connected users	Low
Previous managers	Inferred from history overlap + recommendations text	Heuristic — not reliable	—
Personal interests, extracurriculars, common ground	LLM inference over all sources above	Synthesis, not a new fetch	—

Be realistic: **Instagram** and **previous managers** are the least reliable / highest-risk items on the list. Recommend cutting both from v1.

4. Functional requirements

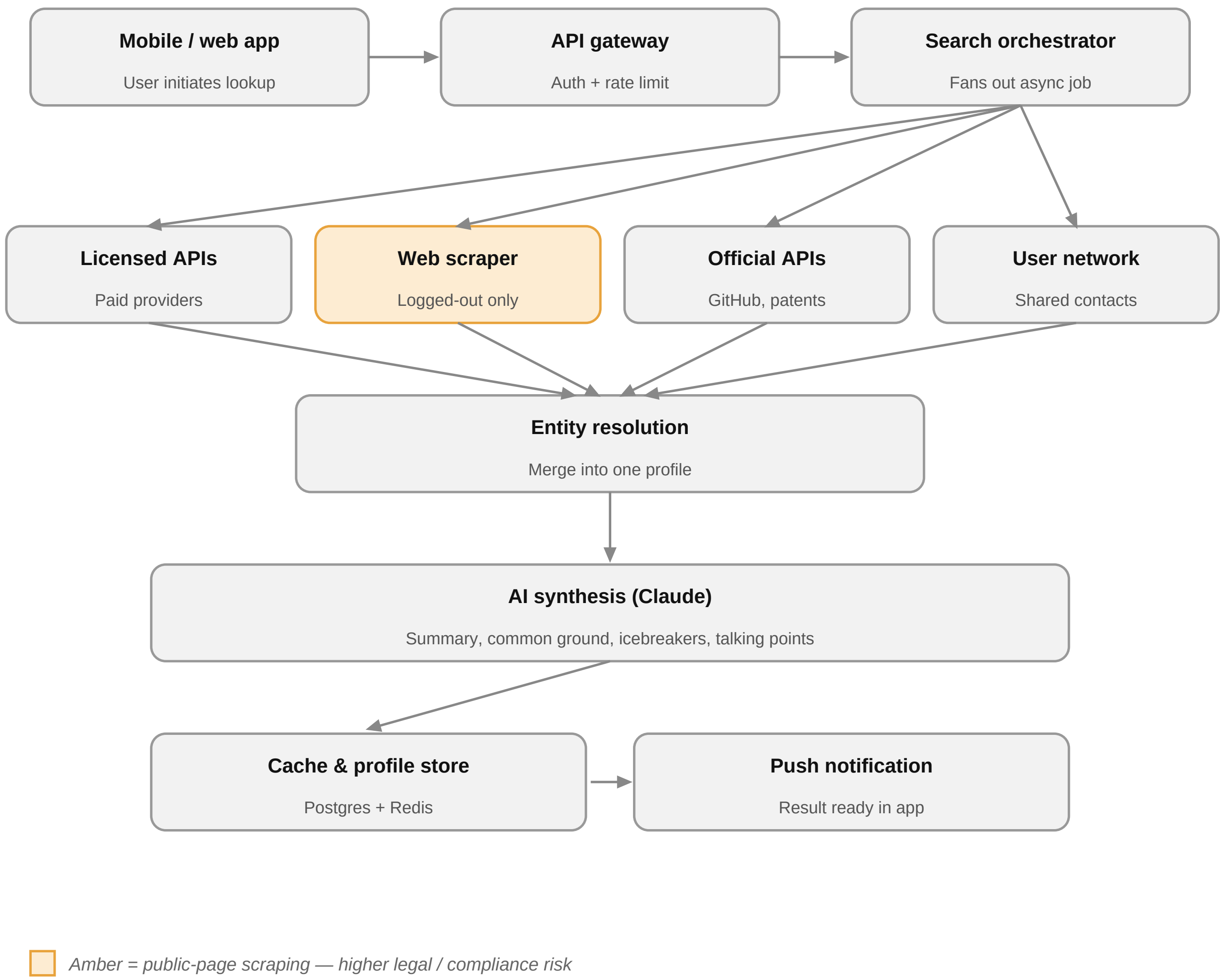
- **FR1 — Auth:** email + OAuth (Google, LinkedIn). LinkedIn OAuth doubles as the legitimate mutual-connections data source.
- **FR2 — Search:** name + optional disambiguators (company, LinkedIn URL, photo/badge/QR scan).
- **FR3 — Disambiguation:** confirm-match UI when multiple people share a name.
- **FR4 — Aggregation:** async fan-out to all connectors, 10–30s cold, cached after.
- **FR5 — Synthesis:** profile summary, common-ground detector (target vs. requesting user's own profile), top-3 talking points, 2–3 icebreaker questions.
- **FR6 — Personal CRM:** post-meeting notes, tags, follow-up reminders.
- **FR7 — Privacy surface:** "claim your profile" opt-out, data-deletion request endpoint, visible data-source disclosure.
- **FR8 — Billing:** subscription tiers, usage metering.

5. Non-functional requirements

- Cold lookup < 20s, cached lookup < 2s.
- Per-source recrawl cadence (e.g. weekly for LinkedIn, daily for news).
- PII encrypted at rest and in transit; strict access logging on every person-profile read — this data is a breach magnet.
- Scraper health monitoring is an ongoing operational workstream, not a one-time build — connectors will break or get blocked regularly.

6. Architecture

Request lifecycle: mobile/web client → API gateway → search orchestrator → four parallel connectors (licensed APIs, web scraper, official APIs, user's synced network) → entity resolution (merge into one profile) → Claude for synthesis → Postgres + Redis cache → push notification back to the app.



Stack recommendation: Node/TypeScript (NestJS) or Python (FastAPI) backend, Postgres + pgvector (interest-similarity matching), Redis, BullMQ/Celery for the async pipeline, Playwright for the scraper connector, React Native for mobile, Claude API for synthesis, Stripe for billing.

7. Legal / compliance risk register

- **LinkedIn will pursue ToS-breach claims** even against logged-out scraping — treat a cease-and-desist as a *when*, not an *if*. Keep connectors swappable so you can fail over to a licensed API quickly.
- **GDPR / CCPA:** you are processing personal data on people who never signed up for the app. You need a lawful-basis assessment, disclosed data sources, and a working erasure pipeline before any EU/CA rollout. Consider geo-restricting lookups on EU/CA data subjects for v1 while validating demand.
- **"People search" is a newly regulated category** in several US states (California's DELETE Act, similar laws in Texas, Oregon, Vermont) — you may need to register as a data broker.
- **App Store risk:** Apple/Google may reject or later pull an app visibly built on scraping a major platform's data without permission — keep a PWA/web fallback distribution channel available.
- Get real legal counsel before scaling past a closed beta — this isn't optional given the category.

8. Monetization

- **Freemium:** N free lookups per month.
- **Individual subscription** (~$15–25/mo) for frequent conference-goers, founders, sales reps.
- **Team tier** for sales orgs — this is where Humanlinker / Crystal Knows monetize today.
- **B2B API licensing** later, once the aggregation + synthesis pipeline is proven.

9. Suggested MVP sequencing

- **1. Legal setup** — privacy policy, ToS, entity/jurisdiction decision.
- **2. Backend + licensed/official connectors only** (GitHub, Scholar, patents, Apollo/PDL) + Claude synthesis — ship something real and legally low-risk fast, validate demand.
- **3. Add the scraper connector** (LinkedIn public pages), isolated so it can be swapped or killed without breaking the product.
- **4. Personal CRM + notes.**
- **5. Billing + mobile polish + badge/QR scan.**
